Module 4 slides: Clearance and Elimination (worked examples)

Module 4 - Drug Elimination and Clearance · Lecture slides, worked examples · 14 Sep 2026

Name _____ Date _____

Five hundred mg of a drug was administered by rapid IV injection. The V_D is 15 L and the elimination half-life is 8 hours. Urine samples were collected for 48 hours and 300 mg of unchanged drug was recovered.

Slide 22 What fraction of the dose is excreted unchanged in the urine?

Slide 22 Calculate k.

Slides 10 and 22

Calculate k_e.

Slide 22 Calculate Cl_T.

Slides 9 and 22

Calculate Cl_R.

Slides 11 and 22

Calculate Cl_H.

Slides 21 and 23

A new antibiotic is actively secreted by the kidney. Its V_D is 25 L in the normal adult and its clearance is 750 mL/min. What is its elimination rate constant in reciprocal hours?

Slides 21 and 23

A new antibiotic is actively secreted by the kidney; V_D is 25 L in the normal adult. The clearance of this drug is 750 mL/min. What is the usual t½ for this drug?

Slides 21 and 23

The same antibiotic has a V_D of 25 L. What would be the new t½ for this drug in an adult with partial renal failure whose clearance of the antibiotic was 150 mL/min?

Slide 17 A patient is 165 cm tall. What is this height in inches?

Slides 16 and 17

A female patient is 165 cm tall. What is her ideal body weight?

Slide 17 Estimate, using the ideal body weight, the creatinine clearance of a 45-year-old female who weighs 61 kg, has a serum creatinine of 1.1 mg/dL, and is 165 cm tall.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Slide 22 Her key

Answer: 0.6 no units

WHAT THE STEM GIVES

D_0	Five hundred mg	the denominator of $\frac{D_u\infty}{D_0}$, 500 mg
V_D	15 L	*not needed: f_e is a ratio of two amounts*
t½	8 hours	*not needed for f_e; it confirms 48 hours is six half-lives*
collection	48 hours	*not needed: it confirms the collection is complete*
$D_u\infty$	300 mg	the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$f_e = \frac{D_u\infty}{D_0}$$

The fraction excreted unchanged is the cumulative amount of unchanged drug appearing in the urine divided by the dose that was given. The 48-hour collection is what makes the 300 mg a cumulative figure rather than a snapshot, so it can stand in for $D_u\infty$.

ALGEBRA

$f_e = 300\ \text{mg} \div 500\ \text{mg}$

Both quantities are amounts in milligrams, so the milligrams cancel and the result is a pure number. The volume of distribution and the half-life are not used in this part; they belong to the later parts of the same question.

ROUND

$f_e = 0.6$, dimensionless

Dr. Mosley asks the class directly what the units of f_e are and confirms there are none, because an amount has been divided by an amount. Reporting 0.6 rather than .6 also satisfies her rule that a number smaller than one must carry a leading zero.

> **DOES THE NUMBER MAKE SENSE?**
> 300 of 500 mg is three fifths, 0.6; a fraction lies between 0 and 1, and 48 hours is six half-lives of 8 hours, so the collection is complete.

Slide 22 Her key

Answer: 0.0866 hr^{-1}

WHAT THE STEM GIVES

D_0	Five hundred mg	*not needed: k comes from the half-life alone*
V_D	15 L	*not needed*

t½	8 hours	the denominator of $\frac{0.693}{t½}$
collection	48 hours	*not needed*
$D_u\infty$	300 mg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{t½}$$

The overall elimination rate constant follows from the half-life for any first-order process, and Dr. Mosley requires it from memory, as she does total clearance, ideal body weight and Cockcroft-Gault. The half-life given is the elimination half-life, so the constant it produces is the overall k.

ALGEBRA

$$k = \frac{0.693}{8}\ \text{hr} = 0.086625\ \text{hr}^{-1}$$

A dimensionless constant divided by a time in hours leaves reciprocal hours, which is the unit a first-order rate constant must carry. The division is carried at full precision here because k feeds the two clearance parts that follow.

ROUND

$$k = 0.0866\ \text{hr}^{-1}$$

Report 0.0866, to four decimal places, the precision she asks for on rate constants so that everyone carries the same value forward. The leading zero is required.

> **DOES THE NUMBER MAKE SENSE?**
> k is 0.693 over 8 hr, a little under a tenth per hour, 0.0866 hr^{-1}; k × 8 must return 0.693.

Slides 10 and 22 Her key

Answer: 0.052 hr^{-1}

WHAT THE STEM GIVES

D_0	Five hundred mg	the denominator of $f_e = \frac{300}{500} = 0.6$
V_D	15 L	*not needed: $k_e = f_e \times k$ uses no volume*
t½	8 hours	gives $k = \frac{0.693}{8} = 0.086625\ \text{hr}^{-1}$
collection	48 hours	*not needed: it confirms the collection is complete*
$D_u\infty$	300 mg	the numerator of f_e

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$f_e = 300\ \text{mg} \div 500\ \text{mg} = 0.6$ and $k = \frac{0.693}{8}\ \text{hr} = 0.086625\ \text{hr}^{-1}$

The excretion rate constant needs both the fraction excreted unchanged and the overall rate constant, so both are assembled first. The fraction comes from the urine collection and the rate constant from the half-life.

SETUP

$k_e = f_e \times k$

Because f_e is the ratio of k_e to k, multiplying the overall rate constant by that fraction recovers the excretion route on its own. This is the only way to obtain k_e here, since no rate constant for a single route can be observed directly from plasma data.

ALGEBRA

$k_e = (0.6)(0.086625\ \text{hr}^{-1}) = 0.051975\ \text{hr}^{-1}$

A dimensionless fraction multiplied by a reciprocal-hour constant leaves reciprocal hours, so k_e has the same units as k. It must be smaller than k, because excretion is only part of the total elimination.

ROUND

$k_e = 0.052\ \text{hr}^{-1}$

Her worked value is 0.052 per hour, rounded from 0.051975. The remaining 0.03465 per hour of the overall constant belongs to the routes other than unchanged excretion.

> **DOES THE NUMBER MAKE SENSE?**
> k_e is the renal share of k: $0.6 \times 0.086625 = 0.052\ \text{hr}^{-1}$, so it must be smaller than k, $0.086625\ \text{hr}^{-1}$, and larger than half of it.

Slide 22 Her key

Answer: 1.3 L/hr

WHAT THE STEM GIVES

D_0	Five hundred mg	*not needed: $Cl_T = k \times V_D$*
V_D	15 L	one factor of $k \times V_D$
t½	8 hours	gives $k = \frac{0.693}{8} = 0.086625\ \text{hr}^{-1}$, the other factor
collection	48 hours	*not needed*
$D_u\infty$	300 mg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_T = k \times V_D$

Total body clearance is the overall elimination rate constant multiplied by the volume of distribution, the relationship Dr. Mosley requires from memory alongside the half-life equation. Both quantities are available here, one from the stated half-life and one stated directly.

ALGEBRA

$$k = \frac{0.693}{8}\ \text{hr} = 0.086625\ \text{hr}^{-1}$$

The half-life is converted to a rate constant first, because clearance needs a fractional rate rather than a time. Reciprocal hours is what the next multiplication requires.

ALGEBRA

$$Cl_T = (0.086625\ \text{hr}^{-1})(15\ \text{L}) = 1.299375\ \text{L/hr}$$

Reciprocal hours multiplied by litres leaves litres per hour, which is the unit a clearance must carry. The 300 mg of urinary drug plays no part in this line; it enters only when the total is split.

ROUND

$$Cl_T = 1.3\ \text{L/hr}$$

Her worked value is 1.3 litres per hour, and she carries that rounded figure into the renal and hepatic parts that follow. Keeping her rounding is what reproduces her 0.78 and 0.52 exactly.

> **DOES THE NUMBER MAKE SENSE?**
> k is 0.693 over 8, 0.086625 hr^{-1}, so a little under a tenth of the 15 L is cleared each hour: a little under one and a half L/hr, and 1.3 L/hr fits.

Slides 9 and 22 Her key

Answer: 0.78 L/hr

WHAT THE STEM GIVES

D_0	Five hundred mg	the denominator of $f_e = \frac{300}{500} = 0.6$
V_D	15 L	with k gives $Cl_T = 0.086625 \times 15 = 1.3$ L/hr
t½	8 hours	gives $k = \frac{0.693}{8} = 0.086625\ \text{hr}^{-1}$
collection	48 hours	*not needed: it confirms the collection is complete*
$D_u\infty$	300 mg	the numerator of f_e

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_R = f_e \times Cl_T$$

Renal clearance is the share of total body clearance taken by the excretion route, and that share is f_e, the fraction excreted unchanged. The relationship $Cl_R = f_e \times Cl_T$ is on the equation sheet; the step that matters is recognising f_e as the multiplier.

ALGEBRA

$$f_e = 300\ \text{mg} \div 500\ \text{mg} = 0.6 \text{ and } Cl_T = (\frac{0.693}{8}\ \text{hr})(15\ \text{L}) = 1.3\ \text{L/hr}$$

Both factors have to be in hand before the product can be formed. The fraction comes from the urine collection and the total clearance from the rate constant and the volume of distribution.

ALGEBRA

$Cl_R = (0.6)(1.3\ L/hr) = 0.78\ L/hr$

A dimensionless fraction multiplied by litres per hour leaves litres per hour, so renal clearance carries the same units as total clearance and must be smaller than it. The arithmetic is six tenths of 1.3.

ROUND

$Cl_R = 0.78\ L/hr$

This is her printed value, obtained from her rounded total clearance of 1.3 L/hr. Carrying the unrounded 1.299375 L/hr instead gives 0.7796 L/hr, which rounds to the same 0.78.

> **DOES THE NUMBER MAKE SENSE?**
> Cl_T is 1.3 L/hr and 0.6 of it is renal: 0.78 L/hr. Cl_R must be less than Cl_T and, with f_e above 0.5, more than half of it.

Slides 11 and 22 Her key

Answer: 0.52 L/hr

WHAT THE STEM GIVES

D_0	Five hundred mg	the denominator of $f_e = \frac{300}{500} = 0.6$
V_D	15 L	with k gives $Cl_T = 0.086625 \times 15 = 1.3$ L/hr
t½	8 hours	gives $k = \frac{0.693}{8} = 0.086625\ hr^{-1}$
collection	48 hours	*not needed: it confirms the collection is complete*
$D_u\infty$	300 mg	the numerator of f_e, which gives $Cl_R = 0.6 \times 1.3 = 0.78$ L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_T = Cl_R + Cl_H$, so $Cl_H = Cl_T - Cl_R$

Clearances add, so the hepatic contribution is whatever is left of the total once the renal contribution is removed. This is the route Dr. Mosley uses, because the liver cannot be sampled the way urine can be collected.

ALGEBRA

$Cl_T = (\frac{0.693}{8}\ hr)(15\ L) = 1.3\ L/hr$ and $Cl_R = (0.6)(1.3\ L/hr) = 0.78\ L/hr$

Both terms in the subtraction have to be built first, the total from k and the volume of distribution, and the renal part from the fraction excreted unchanged. Both are in litres per hour, so they can be subtracted directly.

ALGEBRA

$Cl_H = 1.3\ L/hr - 0.78\ L/hr = 0.52\ L/hr$

Subtracting two quantities in litres per hour leaves litres per hour. The result must be positive and smaller than the total, which is a quick check that the renal part was not computed larger than the whole.

ROUND

$Cl_H = 0.52$ L/hr

Her worked value is 0.52 litres per hour, set up as 1.3 minus 0.78. Using unrounded values throughout gives 0.51975 L/hr, which rounds to the same figure.

> **DOES THE NUMBER MAKE SENSE?**
> Cl_T is 1.3 L/hr and Cl_R is 0.78 L/hr, so the rest, 0.52 L/hr, is hepatic; it is less than the renal part because f_e is above 0.5, and less than Cl_T.

Slides 21 and 23 Her key

Answer: 1.8 hr^{-1}

WHAT THE STEM GIVES

V_D	25 L	the denominator of $\frac{Cl_T}{V_D}$
Cl_T	750 mL/min	becomes 60 × 750 mL/hr = 45 L/hr, the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = k \times V_D, \text{ so } k = \frac{Cl_T}{V_D}$$

Clearance is the rate constant multiplied by the volume of distribution, so dividing the clearance by that volume recovers the rate constant. Total clearance is one of the relationships Dr. Mosley requires from memory, with the half-life, ideal body weight and Cockcroft-Gault.

UNIT

$$Cl_T = 750 \text{ mL/min} \times 60 \text{ min/hr} = 45{,}000 \text{ mL/hr} = 45 \text{ L/hr}$$

The answer is asked for in reciprocal hours, so the clearance has to be expressed per hour rather than per minute before dividing. Multiplying by 60 minutes per hour cancels the minutes, and dividing by 1000 millilitres per litre puts the volume in the same unit as the volume of distribution.

ALGEBRA

$$k = \frac{45 \text{ L/hr}}{25 \text{ L}} = 1.8 \text{ hr}^{-1}$$

Litres per hour divided by litres cancels the litres and leaves reciprocal hours, which is what a first-order rate constant must be. Dividing in the other order would give hours, which is a half-life unit and would signal the error immediately.

ROUND

$k = 1.8$ hr^{-1}

One decimal place is what the two-figure inputs support. A rate constant this large corresponds to a half-life well under an hour, which is consistent with a drug whose clearance of 45 L/hr is far larger than its 25 L volume of distribution.

> **DOES THE NUMBER MAKE SENSE?**

750 mL/min is 45 L/hr, so 45 over 25 L is 1.8 per hour: nearly twice the volume is cleared every hour, a very fast drug with a half-life under half an hour.

Slides 21 and 23 Worked by the drill

Answer: 23.1 min

WHAT THE STEM GIVES

V_D	25 L	the numerator of $\frac{V_D}{Cl_T}$, multiplied by 0.693
Cl_T	750 mL/min	becomes 0.75 L/min, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = 0.693 \frac{V_D}{Cl_T}$$

The slide combines the half-life and clearance relationships into a single expression in the two parameters given here. It avoids computing the rate constant as a separate intermediate, which is the shorter route when clearance and volume of distribution are what the stem supplies.

UNIT

$$Cl_T = 750 \text{ mL/min} = 0.75 \text{ L/min}$$

The volume of distribution is in litres, so the clearance is put in litres per minute by dividing by 1000 millilitres per litre. Keeping minutes as the time unit means the half-life will come out in minutes directly.

ALGEBRA

$$t½ = (0.693)\frac{25 \text{ L}}{0.75 \text{ L/min}} = \frac{17.325}{0.75} = 23.1 \text{ min}$$

The litres cancel and dividing by a volume per minute leaves minutes, which is what a half-life must be. The answer is a time, never a reciprocal time.

ROUND

$$t½ = 23.1 \text{ minutes, which is } 0.385 \text{ hours}$$

One decimal place matches the precision of the inputs. Expressed in hours the same figure is 0.385 hr, and the equivalent route through the rate constant gives 0.693 divided by 1.8 per hour, which is the same 0.385 hr.

DOES THE NUMBER MAKE SENSE?

V_D over Cl_T is 25 over 0.75, a little over half an hour, and 0.693 of that is about 23 min; 23.1 min fits, shorter than V_D over Cl_T.

Slides 21 and 23 Worked by the drill

Answer: 115.5 min

WHAT THE STEM GIVES

V_D	25 L	the numerator of $\frac{V_D}{Cl_T}$, unchanged, multiplied by 0.693
Cl_T	150 mL/min	becomes 0.15 L/min, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = 0.693 \frac{V_D}{Cl_T}, \text{ with } V_D \text{ unchanged at 25 L}$$

Renal failure reduces the clearance but does not alter how the drug distributes between plasma and tissue, so the volume of distribution stays at its normal value. Only the clearance term changes, which is what makes this a one-input comparison.

UNIT

$$Cl_T = 150 \text{ mL/min} = 0.15 \text{ L/min}$$

The clearance is converted to litres per minute so that its volume unit matches the volume of distribution. Keeping minutes gives the half-life in minutes, as in the normal-function case.

ALGEBRA

$$t½ = (0.693)\frac{25 \text{ L}}{0.15 \text{ L/min}} = \frac{17.325}{0.15} = 115.5 \text{ min}$$

The litres cancel and the division leaves minutes. The clearance has fallen to one fifth of its normal 750 mL/min, so the half-life is five times longer, which is the check that the inverse relationship was applied rather than a direct one.

ROUND

$$t½ = 115.5 \text{ minutes, which is } 1.925 \text{ hours}$$

One decimal place matches the inputs. The answer is a time, and it is longer than the normal 23.1 minutes, which is the direction impaired elimination must move the half-life in.

> **DOES THE NUMBER MAKE SENSE?**
> V_D over Cl_T is 25 L over 0.15 L/min, well over two hours, and 0.693 of that is a little under two hours, 115.5 min; 1.925 hours fits.

Slide 17 Her key

Answer: 64.96 inches

WHAT THE STEM GIVES

height	165 cm	divided by 2.54 cm per inch

THE WORKING, ONE LINE AT A TIME

SETUP

$$1 \text{ inch} = 2.54 \text{ cm}$$

The ideal body weight formulas count inches above five feet, so a height given in centimetres is converted first, using 1 inch = 2.54 cm.

UNIT

165 cm ÷ 2.54 cm/inch = 64.96 inches

Dividing centimetres by centimetres per inch cancels the centimetres and leaves inches. Dividing rather than multiplying is what makes the number smaller, which is the expected direction since an inch is larger than a centimetre.

ROUND

64.96 inches, called 65 inches

Her worked value is 64.96 inches, which she rounds to 65 inches for the ideal body weight step. She warns separately that 64.5 inches means sixty-four and a half inches, not sixty-four feet and five inches, because students handle that figure inconsistently.

DOES THE NUMBER MAKE SENSE?

An inch is 2.54 cm, so 165 cm is 165 over 2.54, a little under 65 inches, about 5 ft 5 in; 64.96 fits.

Slides 16 and 17 Her key

Answer: 57 kg

WHAT THE STEM GIVES

height	165 cm	becomes 64.96, taken as 65 inches, so 5 inches over 5 ft
sex	female	selects the 45.5 kg base

THE WORKING, ONE LINE AT A TIME

UNIT

165 cm ÷ 2.54 cm/inch = 64.96 inches, taken as 65 inches

The formula counts inches, so the metric height is converted first and rounded to the nearest inch as she does in her own working. Five feet is 60 inches, so the patient is 5 inches over five feet.

SETUP

Female IBW = 45.5 + 2.3 (inches over 5 ft)

This is the female form of the ideal body weight equation, which she requires from memory because it is not on the equation sheet. The male form starts from 50 rather than 45.5 and uses the same 2.3 per inch, so choosing the wrong starting value shifts the answer by 4.5 kg.

ALGEBRA

IBW = 45.5 + (2.3)(5) = 45.5 + 11.5 = 57 kg

Five inches above five feet multiplied by 2.3 kilograms per inch adds 11.5 kg to the baseline. The result is in kilograms, which is what the creatinine clearance expression requires for the weight term.

ROUND

IBW = 57 kg

Her worked value is 57 kg for this patient. The calculation uses ideal body weight, as the question specifies.

DOES THE NUMBER MAKE SENSE?

165 cm is 65 inches, 5 inches over 5 ft; 2.3 kg per inch adds 11.5 kg to 45.5, so 57 kg, above the 45.5 kg female base.

Slide 17 Her key

Answer: 58.1 mL/min

WHAT THE STEM GIVES

age	45-year-old	in 140 − age = 95
sex	female	the 0.85 factor, and the 45.5 kg IBW base
weight	61 kg	*not needed: the stem asks for the ideal body weight*
S_{Cr}	1.1 mg/dL	in the denominator 72 × 1.1 = 79.2
height	165 cm	becomes 65 inches, so IBW = 45.5 + 2.3 × 5 = 57 kg

THE WORKING, ONE LINE AT A TIME

UNIT

165 cm ÷ 2.54 cm/inch = 64.96 inches, taken as 65 inches, which is 5 inches over 5 ft

The ideal body weight formula counts inches above five feet, so the metric height is converted and rounded as she does in her own working. Sixty inches is five feet, leaving five inches of excess height.

ALGEBRA

$$\text{IBW} = 45.5 + (2.3)(5) = 57 \text{ kg}$$

The female form of the equation is used because the patient is female. The 61 kg actual weight stated in the question is not used, because the question specifies ideal body weight, and she sets all her problems up this way.

SETUP

$$\text{CrCl} = \left[(140 - \text{age})\frac{\text{IBW}}{72 \times S_{Cr}}\right] \times 0.85 \text{ for a female}$$

This is the Cockcroft-Gault equation, which she requires from memory. The 0.85 factor is applied because the patient is female, on the assumption built into the equation that women are smaller and less muscular than men.

ALGEBRA

$$(140 - 45)(57 \text{ kg}) = (95)(57) = 5415$$

The age term subtracts the patient age from 140 and the product is formed with the ideal body weight. Both terms are inputs to an empirical estimate, so the units do not cancel through the expression.

ALGEBRA

$$\frac{5415}{72 \times 1.1 \text{ mg/dL}} = \frac{5415}{79.2} = 68.37$$

The denominator is the fixed constant 72 multiplied by the serum creatinine in milligrams per decilitre. A higher serum creatinine makes the denominator larger and the estimated clearance smaller, which is the direction renal impairment moves in.

ALGEBRA

$$\text{CrCl} = (68.37)(0.85) = 58.11 \text{ mL/min}$$

Applying the female factor reduces the estimate by 15 per cent, on the assumption built into the equation that women carry less muscle for the same height and age. The factor is dimensionless, so the units are unchanged by it.

ROUND

CrCl = 58.1 mL/min

Her worked answer is about 58 millilitres per minute, so 58.1 mL/min is the figure to carry. Creatinine clearance is reported in millilitres per minute, not litres per hour, because it is compared with the normal value of about 120 mL/min.

DOES THE NUMBER MAKE SENSE?

IBW is 57 kg; (140 − 45) × 57 over (72 × 1.1) is 68.37, and the female factor 0.85 brings it to 58.1 mL/min. The female value must be below the unadjusted 68.37.